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#include <iostream>

#include <queue>

#include <string>

using namespace std;

struct Patient {

    string name;

    int age;

    int priority; // 1 = Critical, 2 = Moderate, 3 = Low


    bool operator<(const Patient& p) const {

        return priority > p.priority;

    }

};

int main() {

    priority_queue<Patient> pq;

    int choice;


    do {

        cout << "\n--- Hospital Triage System ---\n";

        cout << "1. Add Patient\n";

        cout << "2. Treat Patient\n";

        cout << "3. Show Next Patient\n";

        cout << "4. Exit\n";

        cout << "Enter choice: ";

        cin >> choice;


        if (choice == 1) {

            Patient p;

            cout << "Enter Name: ";

            cin >> p.name;

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    cout << "Enter Age: ";

    cin >> p.age;

    cout << "Enter Priority (1=Critical, 2=Moderate, 3=Low): ";

    cin >> p.priority;


    pq.push(p);

    cout << "Patient added successfully!\n";

}

else if (choice == 2) {

    if (!pq.empty()) {

        Patient p = pq.top();

        pq.pop();

        cout << "Treating Patient: " << p.name << " (Priority " << p.priority << ")\n";

    } else {

        cout << "No patients in queue.\n";

    }

}

else if (choice == 3) {

    if (!pq.empty()) {

        Patient p = pq.top();

        cout << "Next Patient: " << p.name << " (Priority " << p.priority << ")\n";

    } else {

        cout << "No patients in queue.\n";

    }

}

} while (choice != 4);


return 0;

}

```